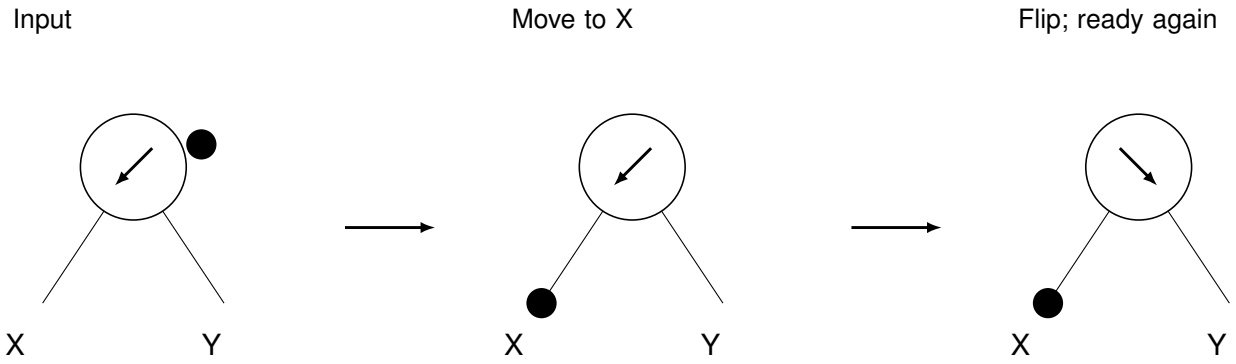
Send counters to the exits, one at a time. At each circle, follow the NEXT arrow, then flip its card. Leave every card as it is for the next counter. Keep the counter beside the card so its arrow stays visible. A partner checks each move.



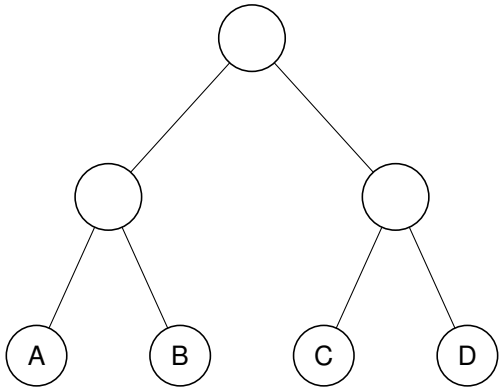
Problem 1: Use the one-switch board. Choose its first arrow. Send 12 counters through and keep their arrival order. Then start a new trial with the other first arrow.

Arrival order

Arrival order

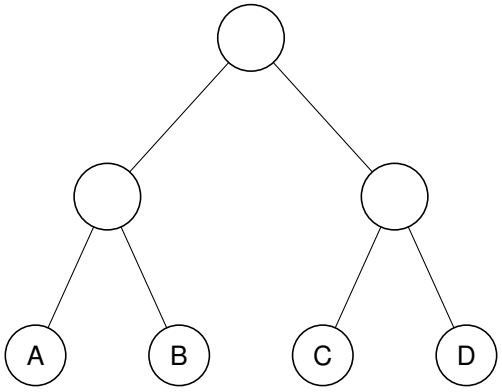
Which arrival strings can this switch make: XYXY, XXYY, YXYX, XYYX?

Problem 2: Use the four-exit tree. Choose starting arrows that send the first counter to D. Keep those cards and send seven more counters. Record the starting arrows and arrival order.

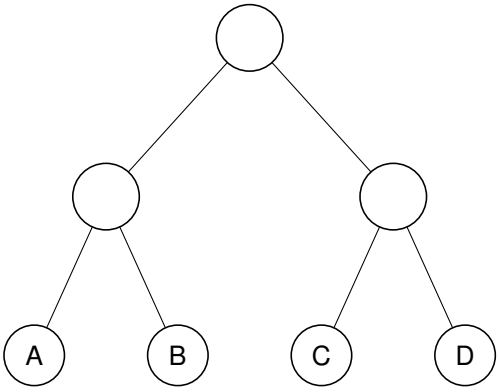


Arrival order

What can you change in the starting arrows while keeping D first? Find every arrival order.



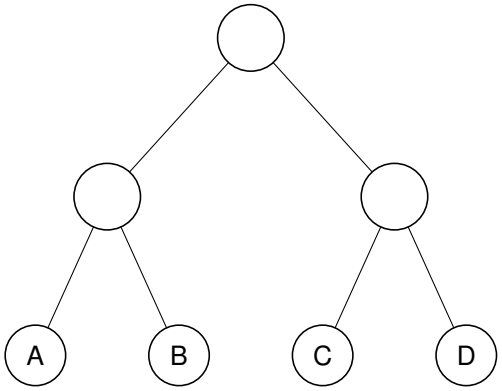
Arrival order



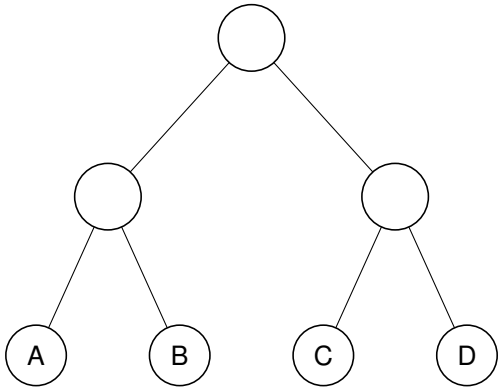
Extra workspace: arrival order

Problem 3: Program the four-exit tree to make each arrival order below. Start a new trial for each order. If an order is impossible, explain what stops it.

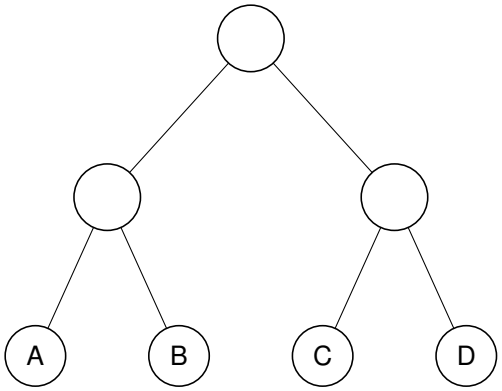
A C B D



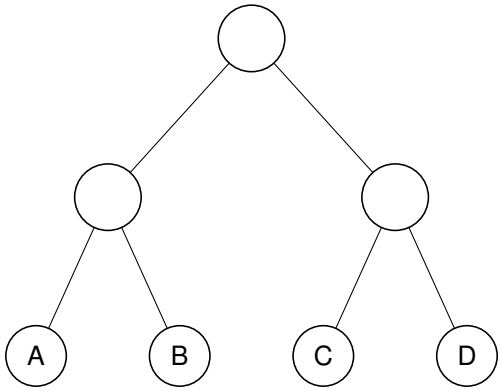
B D A C



A B C D



D A B C



Problem 7: Make a second way to choose exits: toss a coin for each counter, sending heads to X and tails to Y. Compare 12 coin choices with 12 arrivals from the one-switch board.

Coin

Switch

Find a six-counter arrival order that the coin could make but the switch could never make. Could the coin make every order that the switch can make?